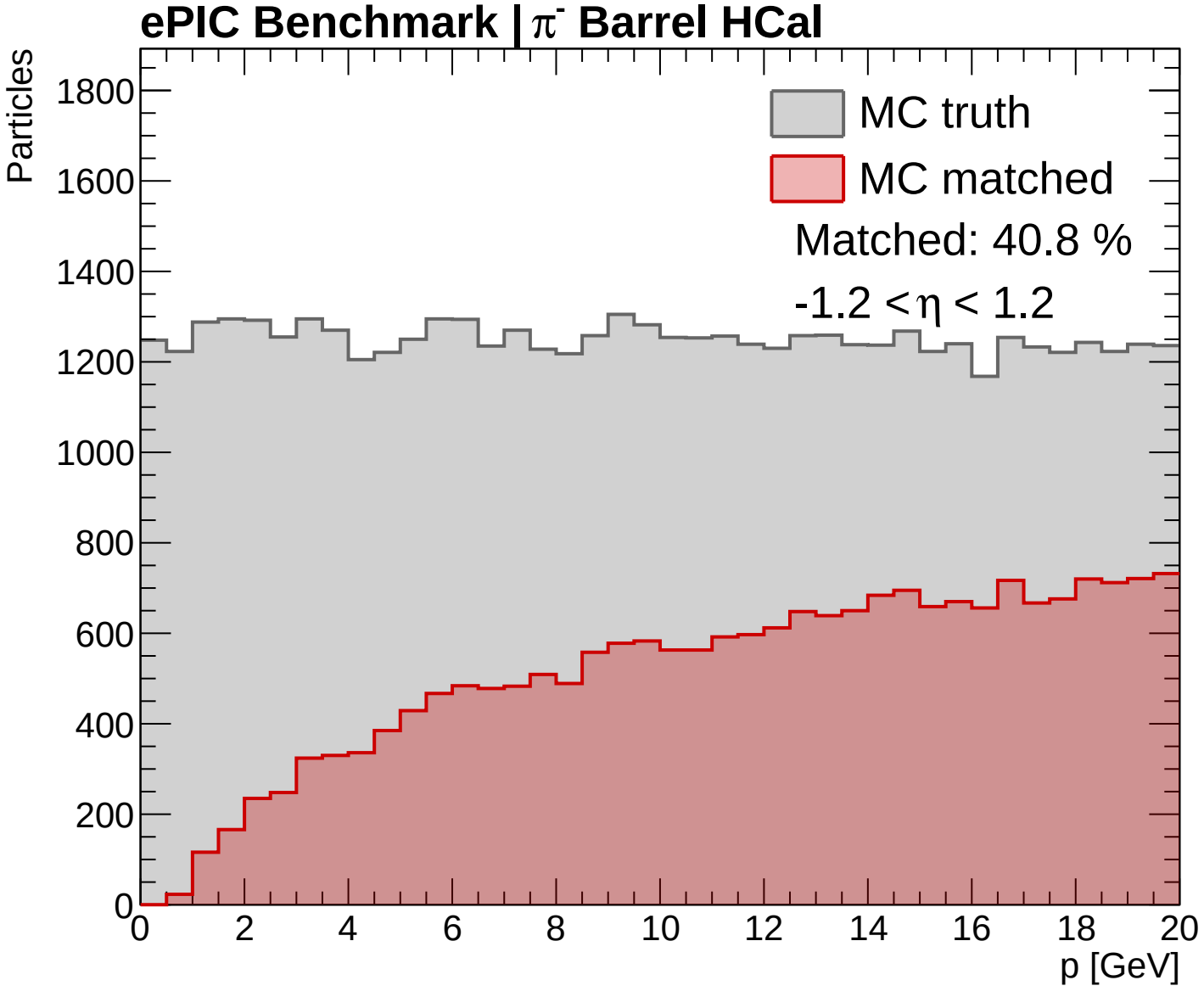
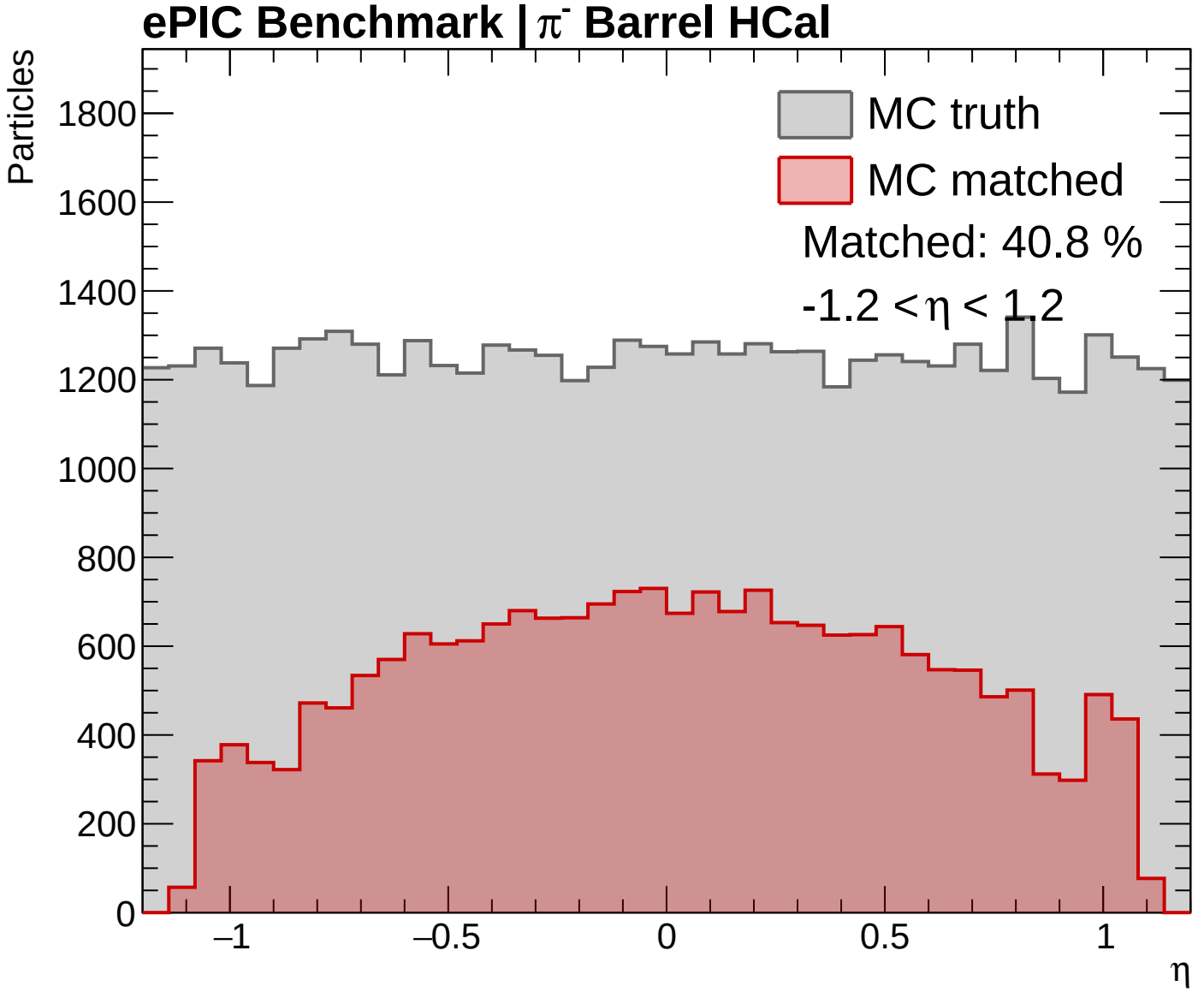
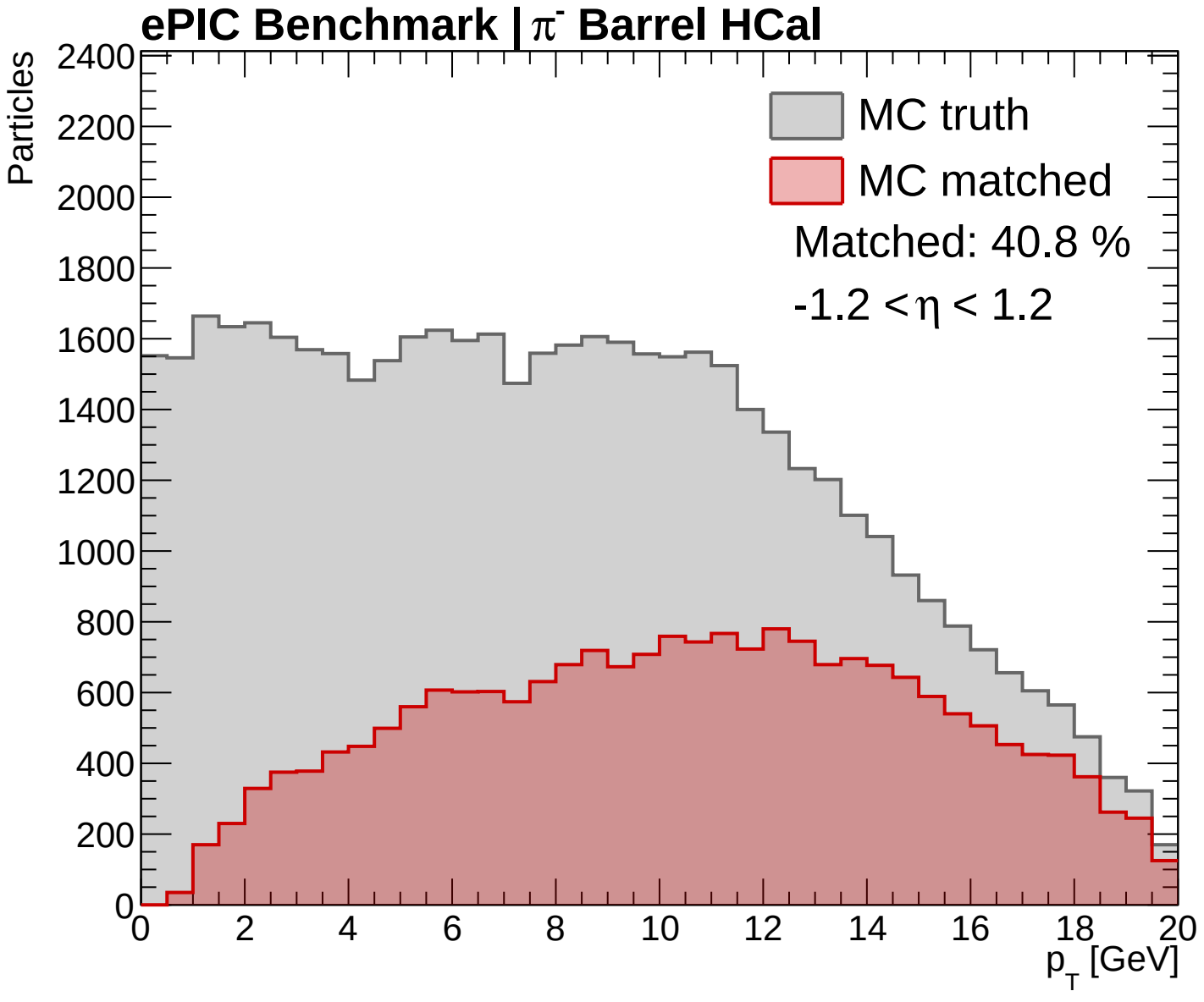
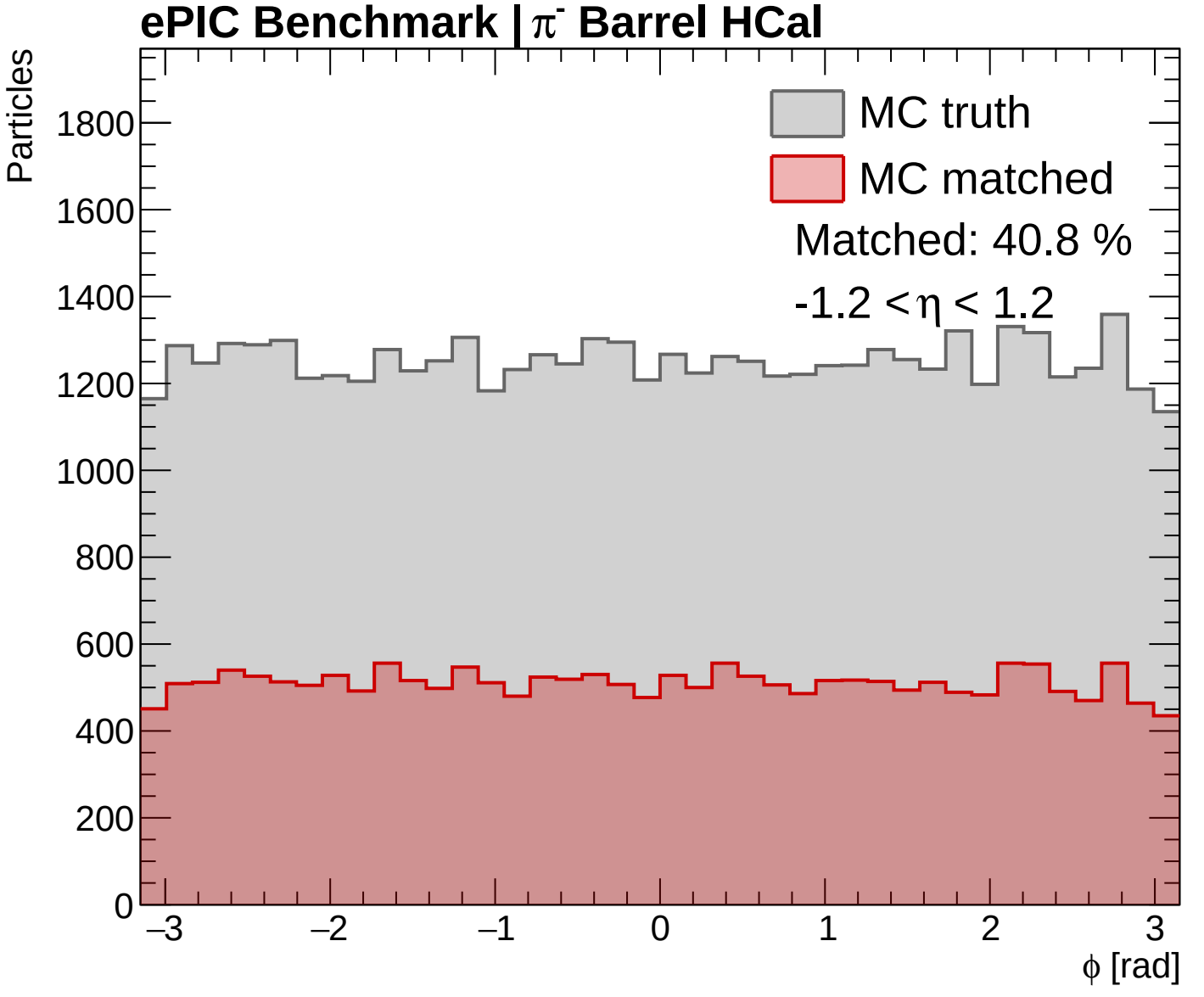
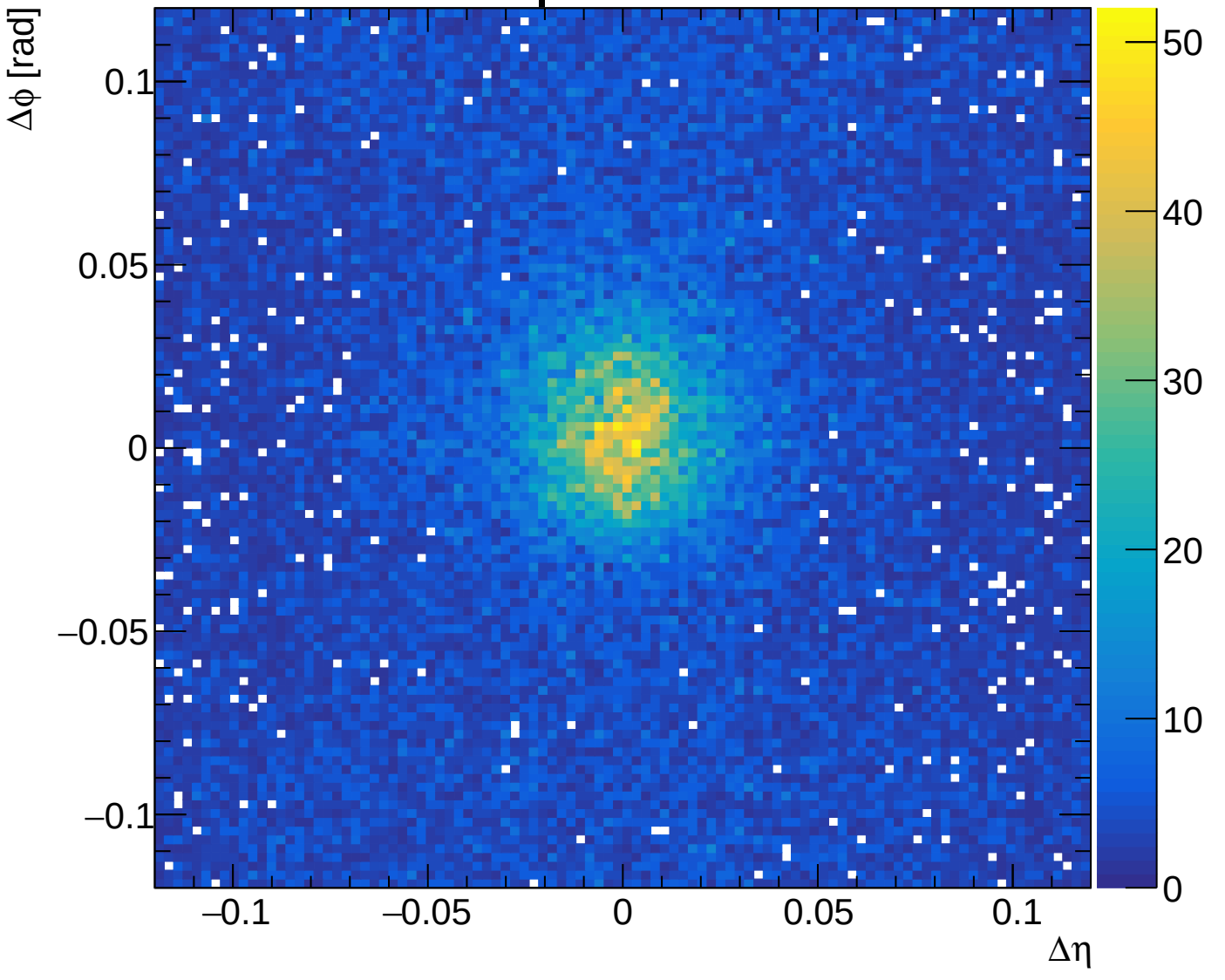
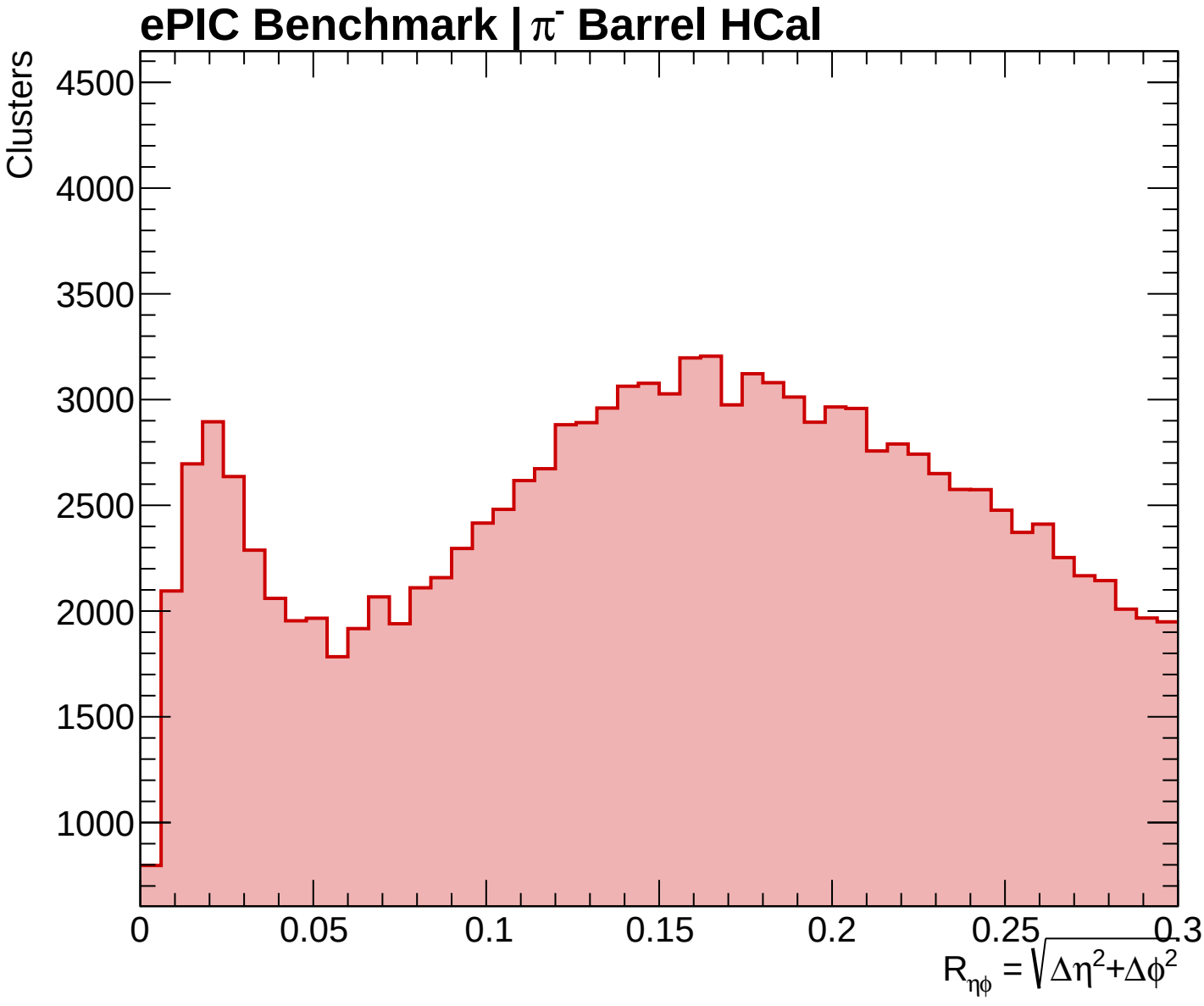

ePIC Benchmark | π⁻ Barrel HCal
Particles
MC truth
MC matched
Matched: 40.8 %
-1.2 < η < 1.2
0
200
400
600
800
1000
1200
1400
1600
1800
0
2
4
6
8
10
12
14
16
18
20
p [GeV]

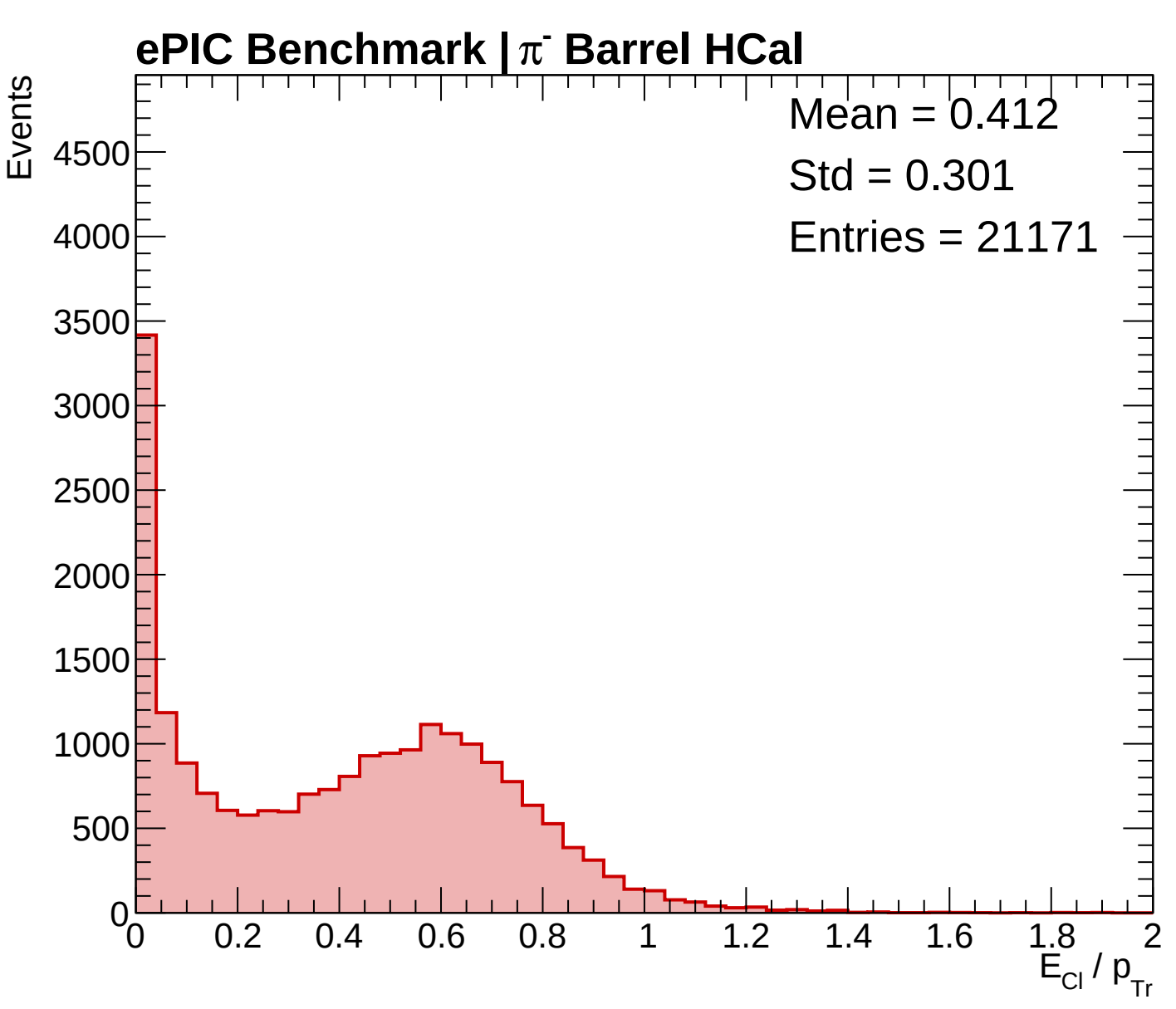

ePIC Benchmark | π⁻ Barrel HCal
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
–1
–0.5
0
0.5
1
η
MC truth
MC matched
Matched: 40.8 %
-1.2 < η < 1.2

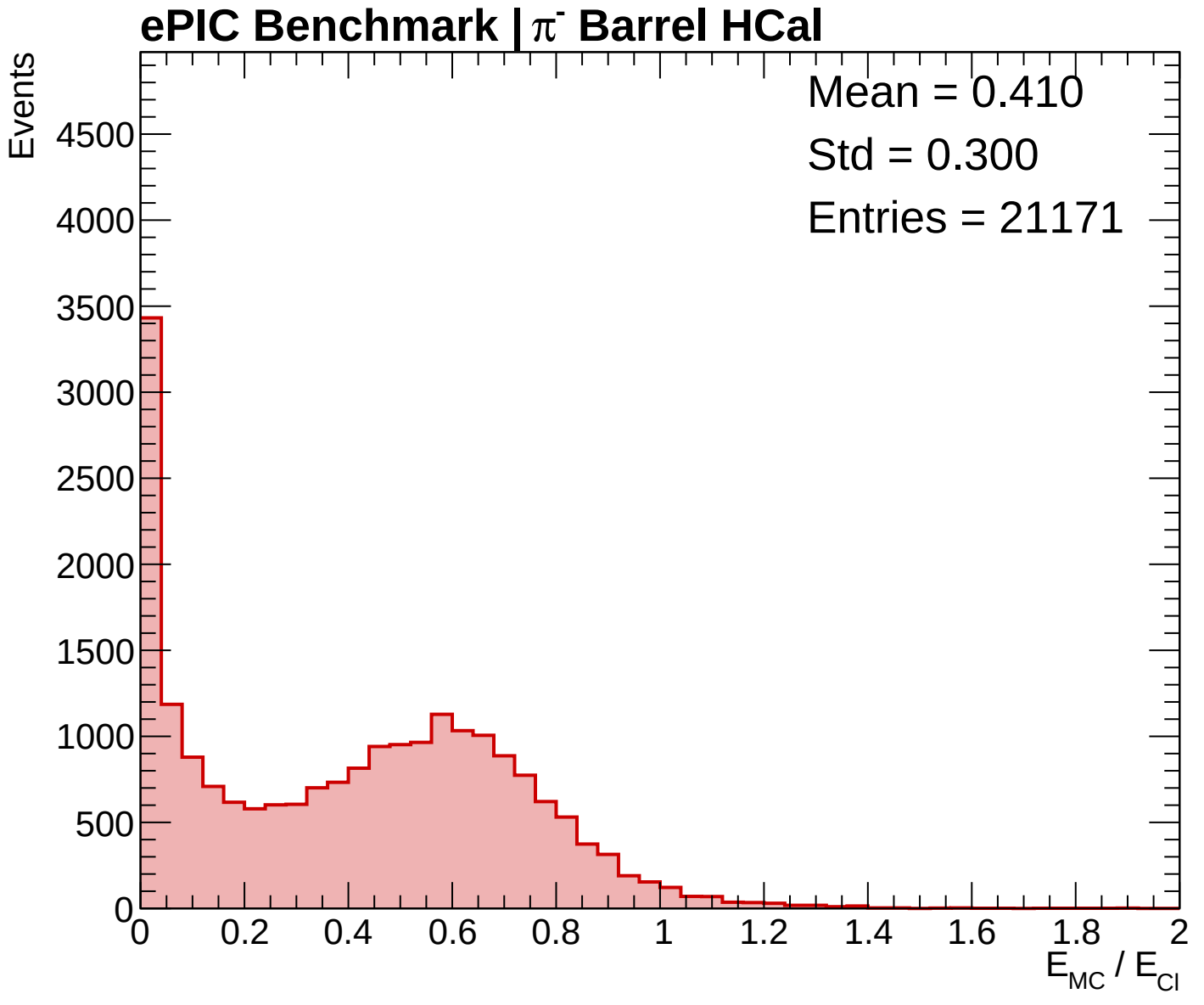

ePIC Benchmark | π⁻ Barrel HCal
Particles
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
2
4
6
8
10
12
14
16
18
20
p_T [GeV]
MC truth
MC matched
Matched: 40.8 %
-1.2 < η < 1.2

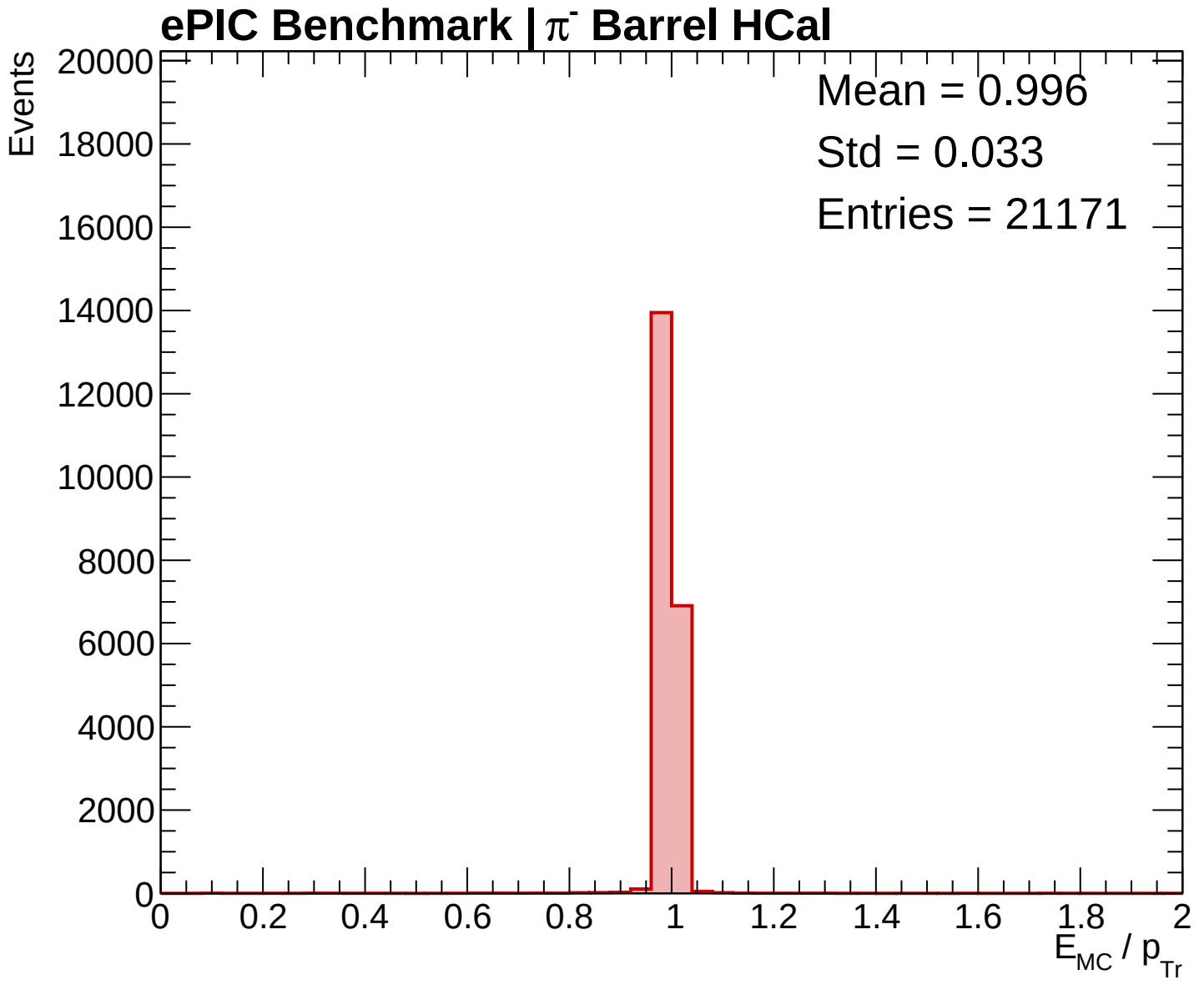

ePIC Benchmark | π⁻ Barrel HCal
Particles
MC truth
MC matched
Matched: 40.8 %
-1.2 < η < 1.2
1800
1600
1400
1200
1000
800
600
400
200
0
–3
–2
–1
0
1
2
3
ϕ [rad]

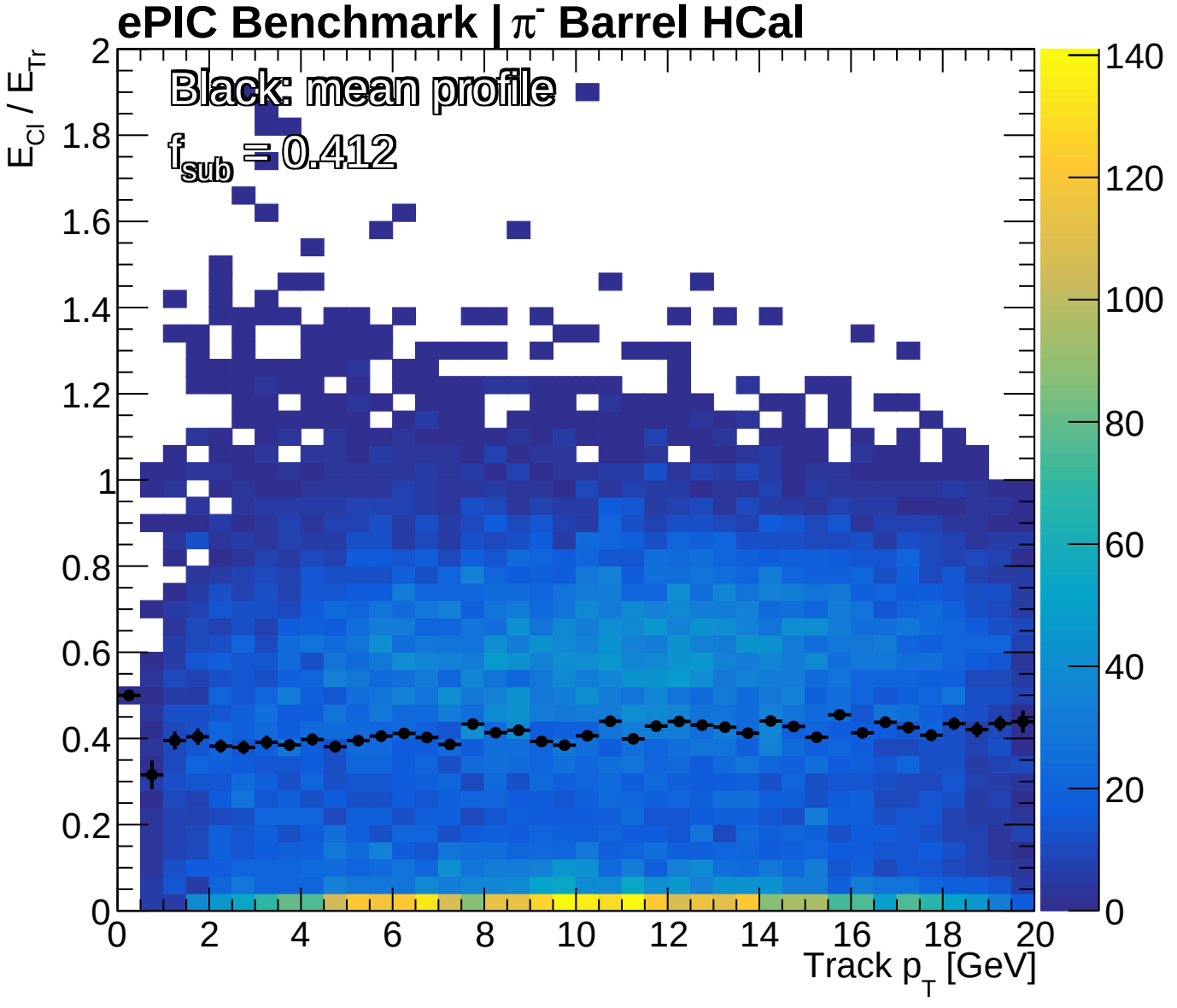

ePIC Benchmark | π⁻ Barrel HCal
Δϕ [rad]
0.1
0.05
0
−0.05
−0.1
−0.1
−0.05
0
0.05
0.1
Δη
50
40
30
20
10
0

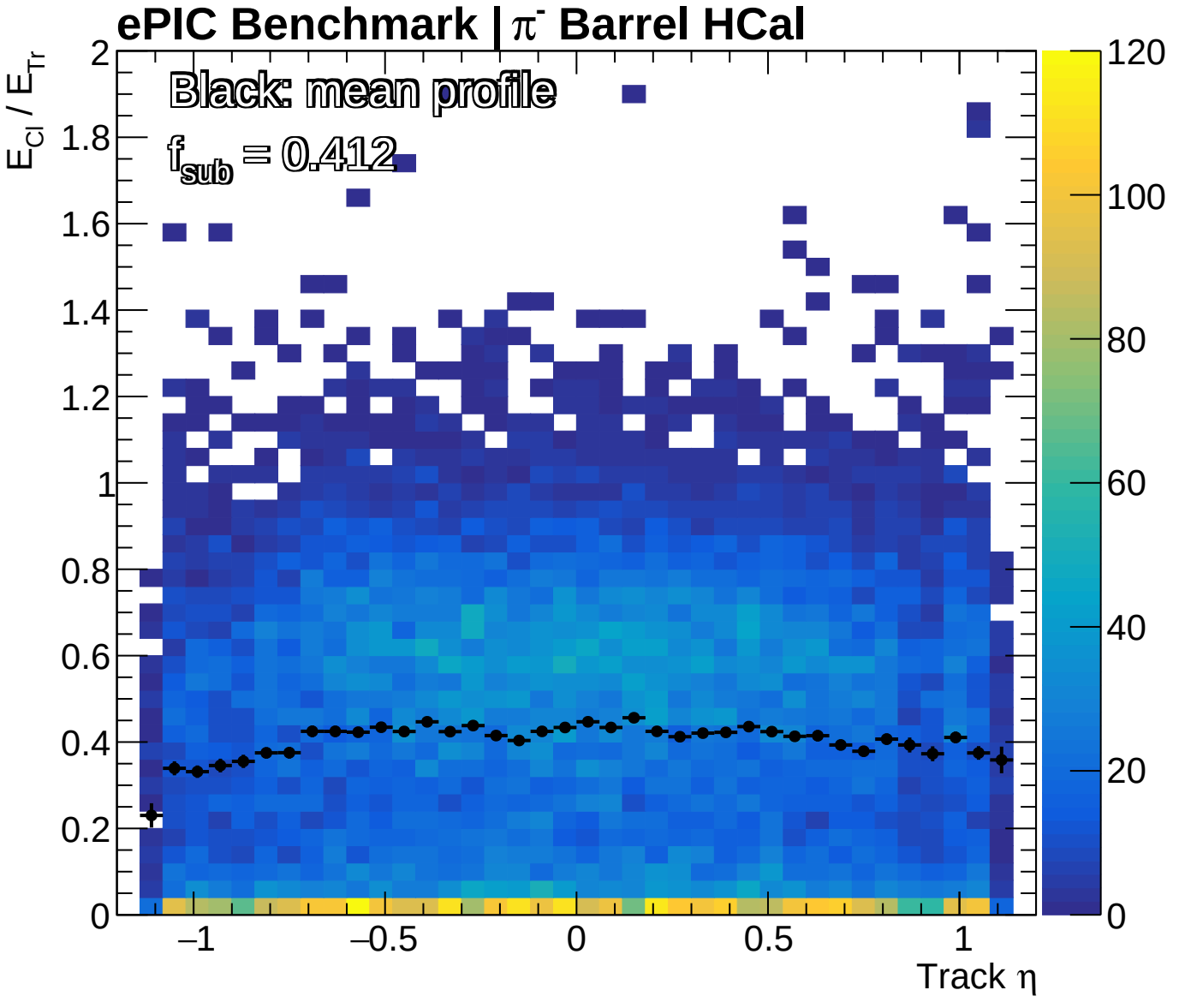

ePIC Benchmark | π⁻ Barrel HCal
Clusters
4500
4000
3500
3000
2500
2000
1500
1000
0
0.05
0.1
0.15
0.2
0.25
0.3
$R_{\eta\phi} = \sqrt{\Delta\eta^2+\Delta\phi^2}$

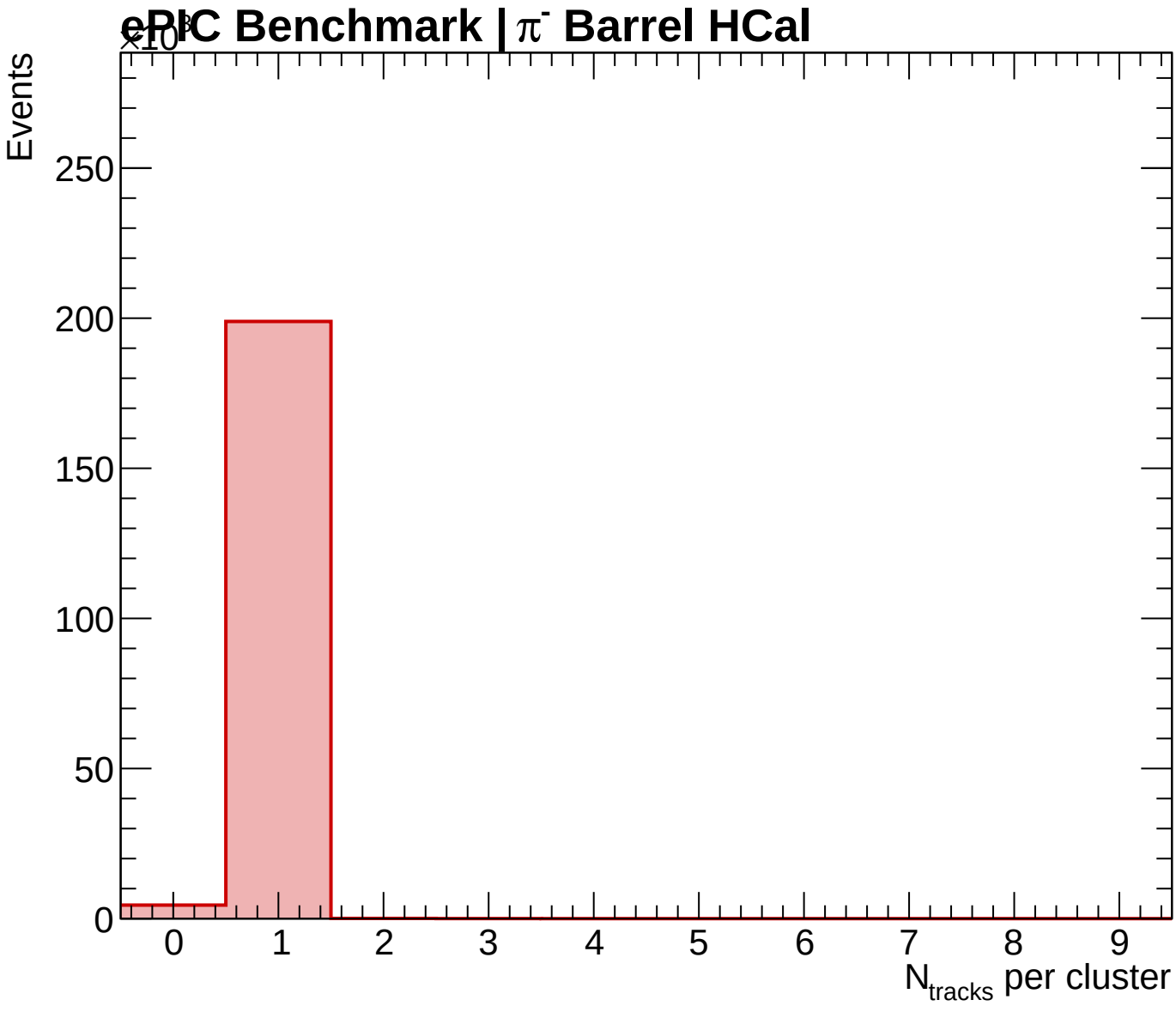

ePIC Benchmark | π⁻ Barrel HCal
Mean = 0.412
Std = 0.301
Entries = 21171
Events
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_Cl / p_Tr

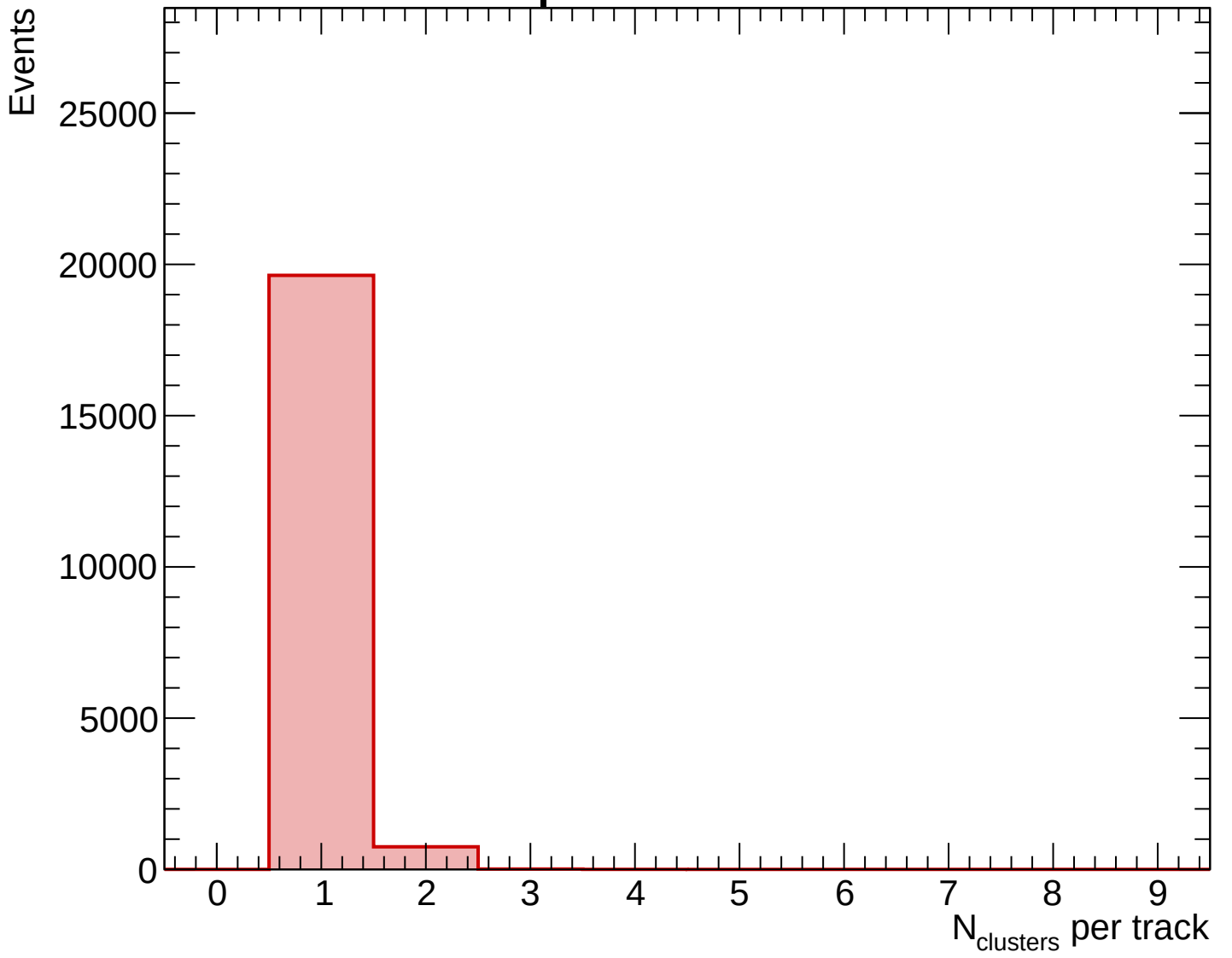

ePIC Benchmark | π⁻ Barrel HCal
Mean = 0.410
Std = 0.300
Entries = 21171
Events
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / E_Cl

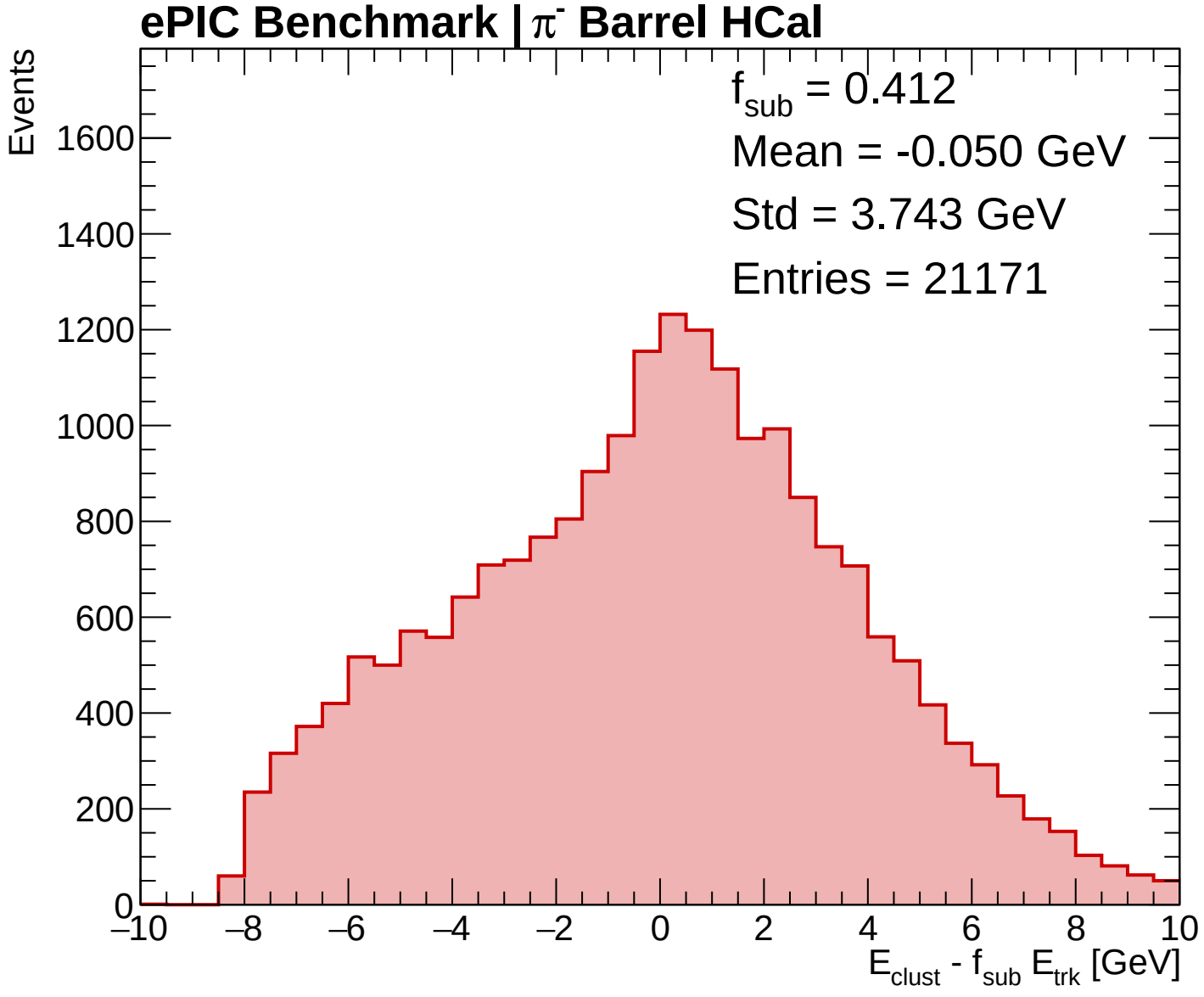

ePIC Benchmark | π⁻ Barrel HCal
Mean = 0.996
Std = 0.033
Entries = 21171
Events
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / p_Tr

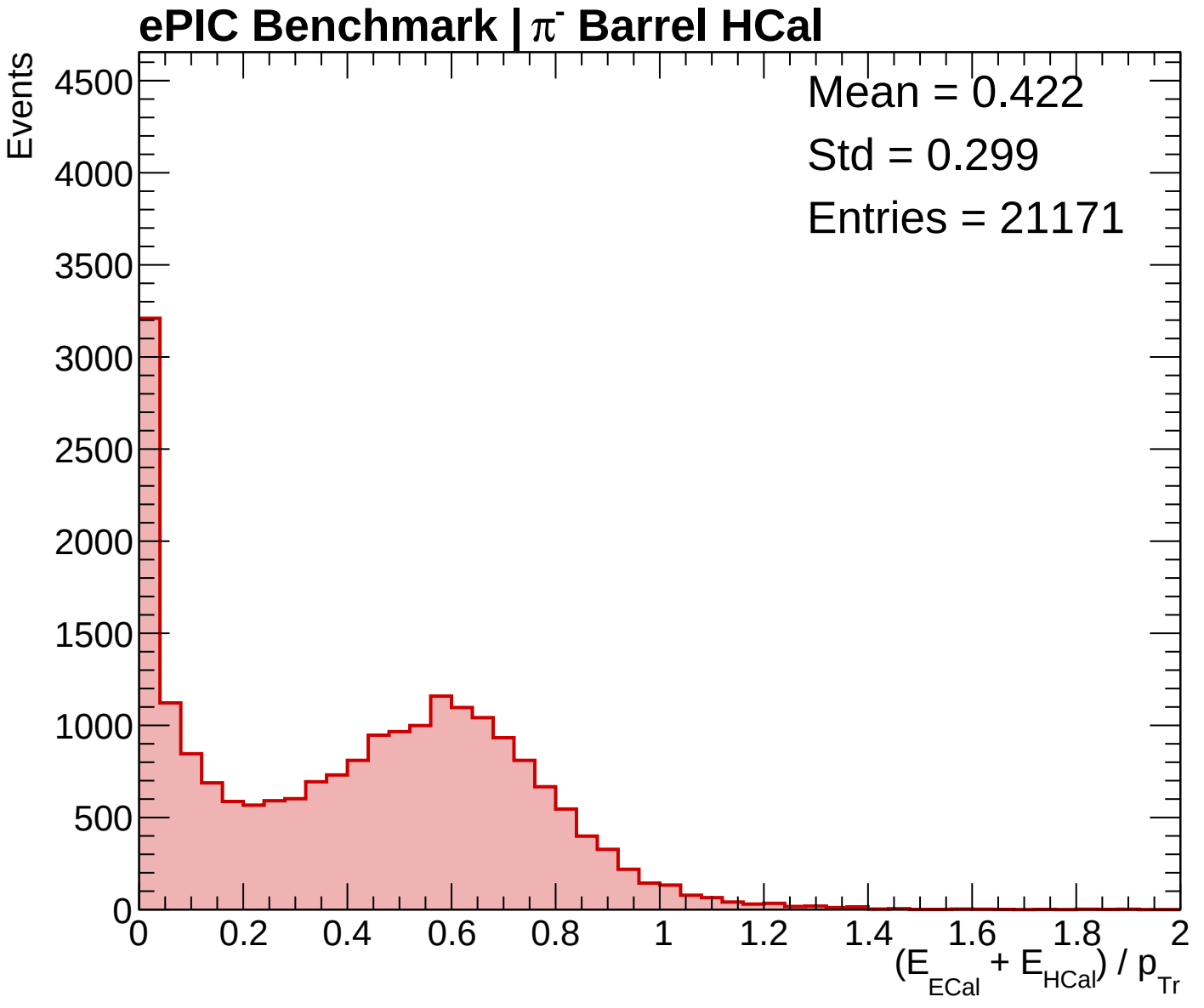

ePIC Benchmark | π⁻ Barrel HCal
Black: mean profile
f_sub = 0.412
E_Cl / E_Tr
Track p_T [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
2
4
6
8
10
12
14
16
18
20
0
20
40
60
80
100
120
140


ePIC Benchmark | π⁻ Barrel HCal
Black: mean profile
f_sub = 0.412
E_Cl / E_Tr
Track η
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
–1
–0.5
0
0.5
1
120
100
80
60
40
20
0


ePIC Benchmark | π- Barrel HCal
×10³
Events
250
200
150
100
50
0
0
1
2
3
4
5
6
7
8
9
N_tracks per cluster


ePIC Benchmark | π⁻ Barrel HCal
Events
25000
20000
15000
10000
5000
0
0
1
2
3
4
5
6
7
8
9
N_clusters per track


ePIC Benchmark | π⁻ Barrel HCal
Events
f_sub = 0.412
Mean = -0.050 GeV
Std = 3.743 GeV
Entries = 21171
0
200
400
600
800
1000
1200
1400
1600
–10
–8
–6
–4
–2
0
2
4
6
8
10
E_clust - f_sub E_trk [GeV]


ePIC Benchmark | π⁻ Barrel HCal
Mean = 0.422
Std = 0.299
Entries = 21171
Events
4500
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_ECal + E_HCal) / p_Tr